Standard Operating Procedure for Resolving Hardware/Environmental Faults on Arista Access Nodes (KOR-ACC-66)

Node: KOR-ACC-66 | **Vendor:** Arista | **Severity:** Minor | **Alarm Code:** ALARM_9782

Description

This SOP outlines the diagnostic and resolution procedures for addressing hardware/environmental faults on Arista Access nodes (KOR-ACC-66) in the Koramangala Datacenter. The failure domain is characterized by ASIC memory parity errors, line card crashes, and power supply redundancy loss. The underlying cause of this failure domain is attributed to the Arista Access node's reliance on a proprietary ASIC (Application-Specific Integrated Circuit) that integrates multiple functions, including packet processing, switching, and memory management. When the ASIC experiences memory parity errors, it can lead to line card crashes, resulting in a loss of redundancy in the power supply. This is exacerbated by the fact that Arista Access nodes are designed to operate in a high-availability configuration, where the loss of one power supply can lead to a system-wide failure. Therefore, it is essential to promptly identify and resolve these issues to maintain network stability and minimize downtime.

Pre-requisite Checks

- check: Verify the Koramangala Datacenter power status and ensure all necessary support personnel are dispatched to the site.

- check: Conduct a redundant link status check to confirm the current configuration and identify potential hotspots for failure.

- check: Collect relevant logs and system information from the affected Arista Access node (KOR-ACC-66) to aid in the diagnostic process.

Potential Root Causes

Cause	Probability
ASIC memory parity error	High
Line card crash	Medium
Power supply redundancy loss	Low

Diagnostic Commands & Expected Outputs

```
Command: show system internal ASIC status, Expected Output: ASIC status: UP | Memory parity errors: 0, Description: This command isolates the ASIC status and memory parity error count, helping to identify if the ASIC is functioning correctly and if memory parity errors are occurring.
```

```
Command: show system internal linecard status, Expected Output: Linecard status: UP | Crash count: 0, Description: This command verifies the linecard status and crash count, helping to determine if the linecard has experienced a crash.

Command: show power redundancy status, Expected Output: Power redundancy status: UP | Redundancy loss: No, Description: This command checks the power redundancy status and verifies if there has been a loss of redundancy.
```

Resolution Steps

Step 1: Step: Power cycle the Arista Access node (KOR-ACC-66), Command: power cycle KOR-ACC-66, Description: Performing a power cycle may resolve issues related to the power supply redundancy loss.

Step 2: Step: Replace the faulty ASIC, Command: replace ASIC KOR-ACC-66, Description: Replacing the faulty ASIC may resolve issues related to ASIC memory parity errors and line card crashes.

Step 3: Step: Verify the linecard status and crash count, Command: show system internal linecard status, Description: Verify that the linecard is functioning correctly and that there have been no crashes.

Post-Check Verification

```
Command: show system internal ASIC status, Expected Output: ASIC status: UP | Memory parity errors: 0, Description: Verify that the ASIC is functioning correctly and there are no memory parity errors.

Command: show system internal linecard status, Expected Output: Linecard status: UP | Crash count: 0, Description: Verify that the linecard is functioning correctly and there have been no crashes.

Command: show power redundancy status, Expected Output: Power redundancy status: UP | Redundancy loss: No, Description: Verify that the power redundancy status is UP and there has been no loss of redundancy.
```

Roll-back Procedures

- Step: Restore the original configuration, Command: rollback configuration KOR-ACC-66, Description: Restore the original configuration to ensure the system returns to its previous state.

- Step: Power cycle the Arista Access node (KOR-ACC-66), Command: power cycle KOR-ACC-66, Description: Perform a power cycle to reset the system.